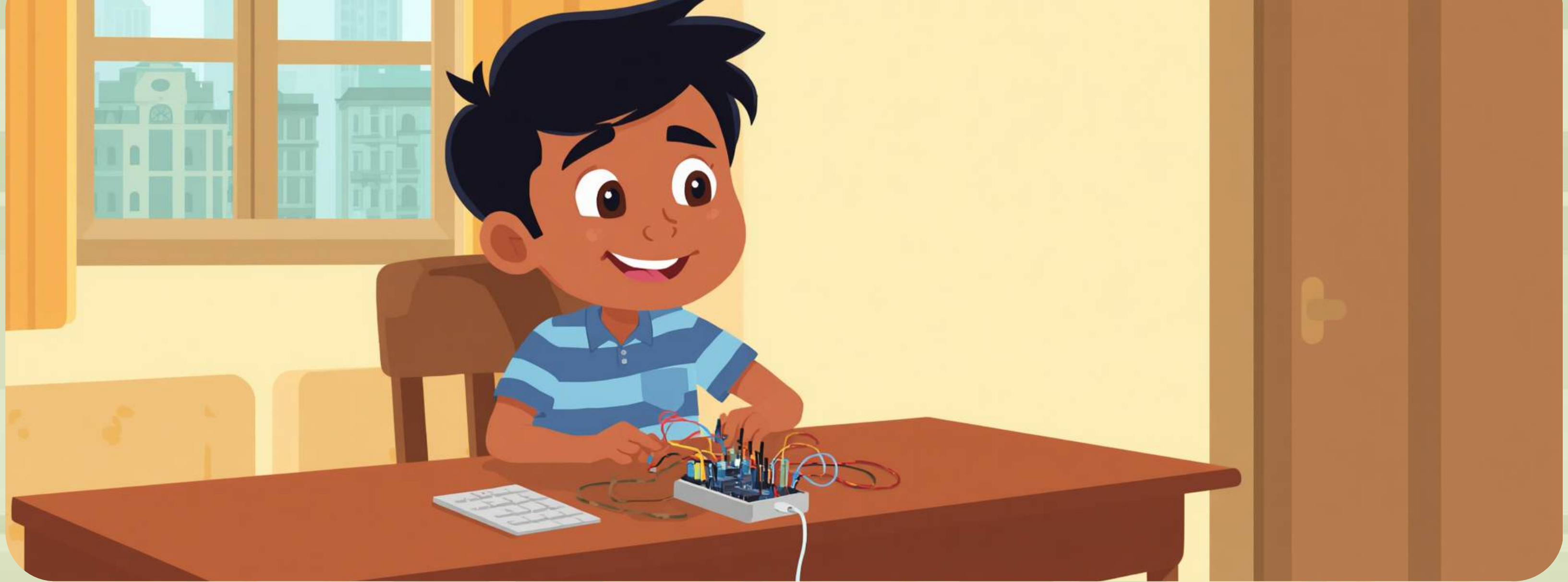


Secret Switches: Logic Gates & Boolean Algebra





Meet Kabir: Our Logic Hero!

From Mumbai 🏠

Kabir wants to build a secret door lock that only opens when conditions are just right! Join him as he discovers the magic of Logic Gates to make his dream invention come true.

What is Boolean Logic?

Logic means deciding if something is True or False!

💡 Light ON = True = 1

💡 Light OFF = False = 0

In Boolean logic, everything is either ON or OFF, YES or NO, 1 or 0. There is no in-between!



True



False

AND Gate: ALL Conditions Must Be Met!

The door opens ONLY if BOTH conditions are TRUE at the same time!



You paid the fee ($A=1$) AND have the permission slip ($B=1$)? Door opens! (Output=1)



If $A=1$ and $B=0$: Output = 0. Missing one condition? The door stays SHUT!



Truth Table: $0 \text{ AND } 0 = 0$ | $1 \text{ AND } 0 = 0$ | $0 \text{ AND } 1 = 0$ | $1 \text{ AND } 1 = 1 \checkmark$



AND Gate shape: D-shape (purple). Input A = Cyan, Input B = Orange, Output = Gold!



OR Gate: AT LEAST ONE Condition!

You get a snack if you have Samosas OR Jalebis!



The OR Gate outputs TRUE if at least ONE input is TRUE. Even one is enough!



Gate Shape: Rocket shape (purple). Inputs: A (Cyan) and B (Orange). Output: Gold!



Truth Table: $0 \text{ OR } 0 = 0$ | $0 \text{ OR } 1 = 1$ | $1 \text{ OR } 0 = 1$ | $1 \text{ OR } 1 = 1$



Real Life: The school gate opens if a teacher OR a student taps their card!



NOT Gate: The Inverter!

Concept: "If electricity is ON, the backup is OFF!"

The NOT gate FLIPS the input — it inverts whatever goes in!

Truth Table:

Input → Output

0 → 1

1 → 0

Gate Shape: Triangle with a bubble (purple)

Input: Cyan | Output: Gold



Salford & Co.

AND Gate

D		$0+1=0$	0	=0
1		$0+1=0$	0	=1
0		$0+1=1$	1	=1
1		$0+1=1$	1	=1



		$0+1=0$	0	=0
15		$0+1=1$	1	=0
16		$0+1=1$	1	=1
17		$0+3=1$	1	=8



		$0+1=0$	0	=0
0		$0+1=0$	0	=0
1		$0+1=0$	0	=0
1		$0+2=0$	0	=0



Truth Tables

➔ **AND | OR | NOT — Side by Side!**

AND: Fee paid + Slip = 1 only when both are

1. OR: Samosa or Jalebi = 1 when any is 1.

NOT: Inverter — 0 becomes 1, 1 becomes 0!



Let's Play! Switch Game

Activity Time!

Try turning switches ON or OFF to see when the bulb lights up! Experiment with AND and OR combinations and discover the magic of logic gates in action.

Circuit Tracing Activity: Kabir's Door Lock!

Kabir's secret door lock uses AND and NOT gates together!

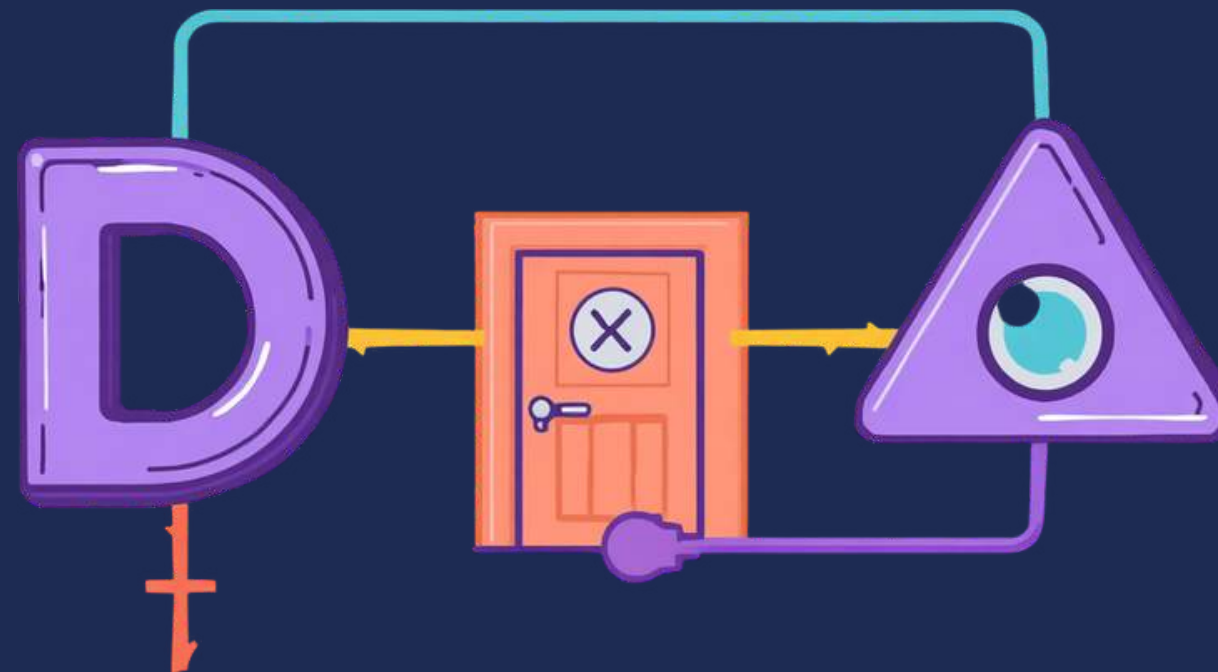
Can you trace the path and decide when the door opens?

● Input A: Key Card Scanned (1 = Yes, 0 = No)

● Input B: Wrong PIN entered (1 = Yes, 0 = No)

NOT Gate flips Input B → then AND Gate checks both!

Door opens only when: $A = 1$ AND $\text{NOT } B = 1$





Ceiling Fan Switch

AND Gate: Fan turns ON only when power is ON AND switch is pressed!



School Bell Rings

OR Gate: Bell rings if it's class time OR break time — either works!



Gully Cricket Umpire

NOT Gate: Umpire signals NO BALL — the opposite of a fair delivery!



UPI Payment Alert

Combination Logic: Payment succeeds AND alarm triggers if fraud detected!

Real Life Logic Around You

Logic gates are everywhere in daily life!



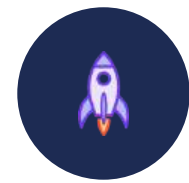
Smart Traffic Signal Project: Delhi

Logic Gates in Real Life

Traffic lights in Delhi change based on sensors using logic gates! AND gates check if a vehicle is present AND the timer is done. OR gates trigger alerts if any lane is blocked. Logic is everywhere around us!



AND Gate: D-shape (purple) — Input A: Cyan, Input B: Orange, Output: Gold. ALL inputs must be 1 to get output 1.



OR Gate: Rocket shape (purple) — Input A: Cyan, Input B: Orange, Output: Gold. AT LEAST ONE input must be 1 for output 1.



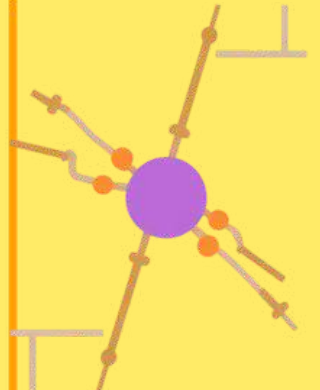
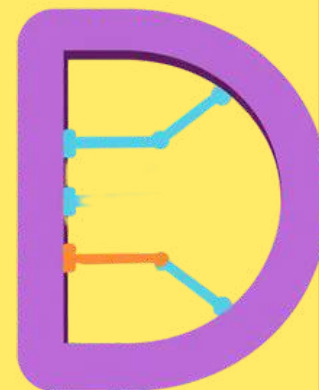
NOT Gate: Triangle with bubble (purple) — Single Input: Cyan, Output: Gold. INVERTS the input: 0 becomes 1, 1 becomes 0!



Color Code Rule: Cyan = Input A, Orange = Input B, Gold = Output, Purple = Gate shape. Use these every time!

Color Codes & Gate Shapes Recap

Remember the shapes, colors, and what each gate does!



Quick Quiz: Test What You Know!

Answer aloud or write it down — let's see what you remember!



Q1: Which gate needs ALL conditions to be true to give output 1?
👉 Answer: AND Gate



BONUS: Kabir's door lock uses which gate to check BOTH conditions at once?
👉 Answer: AND Gate



Q2: Which logic gate is shaped like a rocket?
👉 Answer: OR Gate



Q3: If the NOT gate input is 1, what is the output?
👉 Answer: Output is 0



**Thank You
& Questions!**

